

A Review and Road Map of Entrepreneurial Equity Financing Research: Venture Capital, Corporate Venture Capital, Angel Investment, Crowdfunding, and Accelerators

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Equity financing in entrepreneurship primarily includes venture capital, corporate venture capital, angel investment, crowdfunding, and accelerators. We take stock of venture financing research to date with two main objectives: (a) to integrate, organize, and assess the large and disparate literature on venture financing; and (b) to identify key considerations relevant for the domain of venture financing moving forward. The net effect is that organizing and assessing existing research in venture financing will assist in launching meaningful, theory-driven research as existing funding models evolve and emerging funding models forge new frontiers.

Keywords: *venture capital; corporate venture capital; angel investment; crowdfunding; accelerators; equity financing; entrepreneurship*

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Entrepreneurs pursuing high-growth-potential ventures usually secure financial capital to help fuel their endeavors. While young ventures often rely on a mixture of debt and equity financing, the literature on high-growth-potential start-ups has largely focused on outside equity finance, such as venture capital (VC), corporate venture capital (CVC), angel investment, crowdfunding, and/or accelerators. Across several disciplines, venture financing continues to provide fertile ground for proposing and testing theories within a number of important areas across the individual, organizational, and market levels of analysis.

In light of both the substantial body of research and the emerging trends taking place in the field, much is to be gained from an examination of entrepreneurial equity financing in consideration of future research needs. As such, the first objective of this paper is to organize and review the equity venture financing literature, placing particular emphasis on articles from 2004 forward. Second, we seek to provide a platform from which to pursue new inquiry and future research that facilitates theoretically grounded work in the face of the dynamic entrepreneurial financing landscape. Doing so, we begin by offering an overview of the key funding mechanisms that compose the modern equity funding landscape, followed by a brief review of the earliest phases of venture financing research starting in the early 1980s through 2003. We then focus on the research that has emerged on the different funding mechanisms since 2003. Finally, we give significant attention to future research opportunities associated with each source of funding and their growing interconnectedness.

The Equity Funding Landscape

Entrepreneurial equity investments by venture capitalists, corporate venture capitalists, angel investors, and, more recently, crowdfunders and accelerators represent the key sources of capital that fuels innovation and development. Such investments usually entail significant risk but also have the potential for substantial upside. While equity investors trade capital for a portion of company ownership, several types of equity funding are based on the stage of investment focus, amounts invested, strategic objectives, geographic concentration, and the nature of involvement beyond the provision of capital. Given this distinctiveness, we offer a brief overview of each and then review and assess research on these key forms of equity funding.

First, VC tends to be the most widely recognized form of equity financing, despite funding only a small fraction of start-ups. Venture capitalists raise funds from a set of limited partners (university endowments, pension funds, etc.) and seek to provide a return to these investors through selective investments into a portfolio of young, innovative companies (Gompers & Lerner, 2000). VC firms are typically small, geographically clustered entities, often working closely with the ventures in which they invest to provide guidance and value beyond capital (Sapienza, 1992; Sørensen, 2007). Venture capitalists mainly participate in deals midstage to late stage, recently averaging \$6.4 million per first-time investment (National Venture Capital Association, NVCA, 2016), and continue drifting to larger and later-stage investments (Hellmann & Thiele, 2015). Because venture capitalists provide returns to their limited partners within 10 or so years, there is often a focus on realizing timely exits via an acquisition or initial public offering (IPO). While exhibiting cyclical ebbs and flows since arising in the 1960s, venture capitalists are currently averaging investments of about \$32 billion annually (NVCA, 2016).

Second, CVC denotes the systematic practice by established corporations of making equity investments in entrepreneurial ventures. CVC is distinct from traditional VC in that funds are invested by arms of corporations as extensions of their primary focus (e.g., Google and Samsung). Corporate investors are usually looking to bring long-term value to their firms and focus more on early stage to midstage ventures. During the latest cyclical uptick, CVC arms have grown from 625 to 1,100 between 2010 and 2014 (VentureBeat, 2014) and provided \$7.7 billion into 930 rounds in 2015 (NVCA, 2016). These corporate investors have had a significant impact on investee companies by providing capital, complementary assets, shared industry knowledge, and access to customers as well as in shaping innovation strategies for their own firms.

Third, angel investors are accredited individuals who invest their own personal capital into young ventures. Angels are often former entrepreneurs who seek to fund and add value/guidance to investee firms in their area of expertise. Such individuals tend to take a less formal approach to investing, particularly with regard to the level of due diligence conducted and the formality of contracts and control involved. Most recently, angels have heightened their impact by forming angel investor groups (Kerr, Lerner, & Schoar, 2014) and by providing platforms, both online and in person, for individual angels to evaluate and invest in high-potential deal flow collectively (e.g., Tech Coast Angels). While historically a highly independent and fragmented market, the angel investing landscape is trending toward more centralized angel networks and groups across the globe. Often investing at the earliest stages of the venture life cycle, angels were estimated to have invested \$25 billion into over 70,000 young ventures in 2015 (Sohl, 2015).

Finally, and most recently, crowdfunding and accelerators are two emerging forms of equity funding mechanisms. Equity crowdfunding is where a large volume of online investors contribute smaller amounts for fractions of company ownership (Vulkan, Åstebro, & Sierra, 2016). While equity crowdfunding initially faced significant legal challenges, it is experiencing rapid growth as legal barriers are being relaxed in many countries (Ahlers, Cumming, Günther, & Schweizer, 2015). Accelerators are cohort-based programs that trade a configuration of mentorship, work space, and/or funding, often in exchange for equity. The capital provided typically runs from \$25,000 to \$150,000 and is offered at the very earliest stages. At its core, entrepreneurs apply for an opportunity to develop (or “accelerate”) their concepts on site during a fixed time period (3–6 months). Those accepted are provided with an immersive experience that equips entrepreneurs with many of the essential tools required to succeed (Hathaway, 2016). Cohorts conclude with a “demo day,” where concepts are pitched to potential investors and stakeholders. Accelerators (e.g., Y-Combinator) are proliferating across the globe, increasing from 1 in 2005 to over 500 in 2015 (Shane, 2016).

Together, the above represents the modern entrepreneurial equity funding landscape. We next turn to reviewing the growing body of academic literature that has evolved in this space.

Review Framework and Method

To review venture financing research, we searched for articles focusing on VC, CVC, and angel investment. While equity crowdfunding and accelerators are also of recently growing interest, there is not enough research at this juncture to constitute a meaningful review; we address these in the future research section. The time frame for this study is published

research from 1980 to 2016. We restricted the boundaries of our word search to the title and abstract for specific key words. For VC, we searched for the terms “venture capital,” “venture capitalist(s)” and “VC.” For CVC, we searched “corporate venture capital” and “CVC.” For angel investment, we searched for “angel” and “informal venture capital.”

We searched for the relevant articles in leading management, entrepreneurship, finance, and sociology journals. These included *Academy of Management Journal (AMJ)*, *Academy of Management Review (AMR)*, *Administrative Science Quarterly (ASQ)*, *American Journal of Sociology (AJS)*, *American Sociological Review (ASR)*, *Entrepreneurship Theory and Practice (ETP)*, *Journal of Business Venturing (JBV)*, *The Journal of Finance (JOF)*, *Journal of Financial Economics (JFE)*, *Journal of Management (JOM)*, *Journal of Management Studies (JMS)*, *Management Science (MS)*, *Organization Science (OS)*, *The Review of Financial Studies (RFS)*, *Strategic Entrepreneurship Journal (SEJ)*, and *Strategic Management Journal (SMJ)*. Thus, our analysis of venture financing research includes an interdisciplinary set of 16 leading journals.

Our search yielded 418 article classifications. Given the size and timespan of the paper set, and the variation in equity funding mechanisms, we divided the articles in three key ways. First, greater weight is placed on the more recent research. In doing so, we divided the 418 articles into two tranches—1980 to 2003 and 2004 to 2016. The 1980-to-2003 block covers the venture financing field during its time of emergence. We focus the majority of our attention on articles from 2004 forward. This allows for a more thorough assessment of the current trends that began taking shape in the early 2000s. Second, in offering a more in-depth review of articles published from 2004 forward, we divided the review sections by funding models of VC, CVC, and angel. Third, nested within each funding mechanism, or section, we classified articles across three units of analysis: The *individual level*, the *organizational level*, and the *market level* (see Table 1).

Venture Financing Research: 1980 to 2003

The modern landscape for entrepreneurial finance began emerging in the 1950s and 1960s as new technologies became available for commercialization and entrepreneurs started pursuing high-growth-potential ventures. As equity funding started to gain a foundation by the early 1980s, scholars began to take note and started to probe the nature of equity financing and how it differed from the more traditional sources of capital. Over the last several decades, this research has become increasingly prevalent and sophisticated, regularly appearing in a set of leading journals that span several disciplines. This section provides an overview of the earlier stages of this work by highlighting many of the impactful articles that appeared in the period from 1980 to 2003. Notably, aside from a small handful of CVC and angel investment articles, research on VC composes the vast majority of this early research—thus constituting the majority of our focus.

Early articles on VC were often descriptive of how the process works, the role of key players, and some of the important concepts and frameworks that proved foundational for this developing stream of research (e.g., Bygrave, 1988; Elango, Fried, Hisrich, & Polonchek, 1995; Florida & Kenney, 1988; Gorman & Sahlman, 1989; Robinson, 1988). Most of this early research relied heavily on primary field research complemented by Venture Economics' secondary data. Tyebjee and Bruno (1984) characterized VC deals as being methodical and identified five sequential steps of deal origination, deal screening, deal evaluation, deal structuring,

Table 1
Summary of Venture Capital, Corporate Venture Capital, and Angel Research

	Primary Contributions	Sample Studies at Each Level	Paper Classification Count
VC			
Individual	Investors' evaluation of prospective deals and the subjective judgment in VC decision-making processes The entrepreneur's decision to seek VC Relationship between venture capitalists and the entrepreneurs (i.e., entrepreneur-venture capitalist dyad)	Dimov & Shepherd (2005) MacMillan, Siegel, & Narasimha (1986) Franke, Gruber, Harhoff, & Henkel (2006) Drover, Wood, & Zacharakis (in press) Murnieks, Haynie, Wiltbank, & Harting (2011)	47
Organizational	Alleviating problems with asymmetric information in financial markets The value added by VC and performance outcomes for entrepreneurial firms Mitigating risk and contact mechanisms (e.g., multistage investment vehicle mechanisms, stock options, covenants, types of securities)	Sahlman (1990) Megginson & Weiss (1991) Gompers (1995) Stuart, Hoang, & Hybels (1999) Hochberg, Ljungqvist, & Lu (2007) Kaplan & Strömberg (2004) Hsu (2004) Baum & Silverman (2004) Sørensen (2007)	201
Market	How macrolevel factors shape actions and outcomes (e.g., market entry, screening processes, syndication patterns, financial performance) The role of government incentives, formal institutions, and informal institutions in shaping country-level differences in VC markets	Inderst & Müller (2004) Ahlstrom & Bruton (2006) S. H. Lee, Peng, & Barney (2007) Cumming, Schmidt, & Walz (2010)	93
CVC			
Individual	CVC personnel and mind-set Individual compensation of CVC decision makers	Dushnitsky & Shapira (2010) Dokko & Gaba (2012)	5

(continued)

Table 1 (continued)

	Primary Contributions	Sample Studies at Each Level	Paper Classification Count
Organizational	Economic and behavioral antecedents of CVC The CVC unit within a given firm CVC performance outcomes Considerations and performance outcomes from the entrepreneurial firm's perspective	Dushnitsky & Lenox (2005a) Park & Steensma (2012) Benson & Ziedonis (2010) Kanter, North, Bernstein, & Williamson (1990)	35
Market	Trends in CVC investment activity CVC as part of broader changes in R&D strategies	Chemmanur, Loutskina, & Tian (2014) Sahaym, Steensma, & Barden (2010)	7
Angel Investment Individual	Individual angel investor's decisions to invest Motivations for entrepreneurs to pursue angel capital	Freear, Sohl, & Wetzel (1994) Maxwell, Jeffrey, & Lévesque (2011) Maxwell & Lévesque (2014)	16
Organizational	Decision making within angel groups Value added and performance of entrepreneurial firms receiving angel capital	Carpentier & Suret (2015) Kerr, Lerner, & Schoar (2014)	10
Market	Comparing and contrasting angel markets with other mechanisms (e.g., VC)	Hellmann & Thiele (2015) Carpentier, Lerner, & Suret (2010)	8

Note: VC = venture capital; CVC = corporate venture capital.

and postinvestment activities. They also modeled expected returns based on market attractiveness and product differentiation as well as perceived risk based primarily on the capabilities of the management team and environmental threats.

Both MacMillan, Siegel, and Narasimha (1986) and MacMillan, Zemann, and Subbanarasimha (1987) also made some important early contributions by identifying the criteria that venture capitalists use in evaluating prospective investments to distinguish between successful and unsuccessful ventures. MacMillan, Kulow, and Khoylian (1989) examined how venture capitalists remain involved in their funded investments. Similarly, Timmons and Bygrave (1986) highlighted the nonfinancial or managerial contributions that venture capitalists bring to the funded ventures, demonstrating that backing a venture was just as much or more about managing and providing valuable input to the entrepreneurs as about providing financial capital. In a complementary article, Gorman and Sahlman (1989) asked, "What do venture capitalists do?" and noted among other things that venture capitalists spend about half of their time monitoring approximately nine portfolio ventures. In the most-cited VC article from this era, Sahlman (1990) described the structure and governance of venture capitalists. Given the fledgling nature of VC organizations, they were compared and contrasted with mainline public corporations as well as with leveraged buyout organizations. Barry, Muscarella, Peavy, and Vetsuypens (1990) also brought attention to the specialized investments of venture capitalists, highlighting their role in shaping and governing funded ventures. These articles delineated how venture capitalists approach the potential funding of a new venture and how they help develop the ventures in which they invest.

Linkages were also made between VC funding and economic growth. Timmons and Bygrave (1986) characterized VC investments as encouraging high-technology ventures, suggesting that such investments were an engine of innovation and high growth in the United States. Another important article with market-level implications was Bygrave's (1987) work on syndicated investments by venture capitalists. This work brought significant attention to the extensive networks that venture capitalists use to share information on potential deals as well as to manage their financial risk by coinvesting.

Early work on CVC also tended initially to be descriptive. Winters and Murfin (1988) offer a general survey of the CVC industry and an overview of why and how corporations might benefit from CVC. Rind (1981) describes the evolution and state of CVC and discusses the implications associated with several types of CVC strategies: new ventures division, wholly owned ventures, new-style joint ventures, direct VC, and outside partnerships. Additional studies also explored various approaches and outcomes of CVC investment activity, recommending solutions for improving investment success (Siegel, Siegel, & MacMillan, 1988). Early research on angel investment follows a similar path by covering the general angel market; discussing its scale, efficiency, and patterns (Harrison & Mason, 1992; Wetzel, 1987); and profiling investor attributes, attitudes, and behaviors (Aram, 1989; Fiet, 1996; Freear, Sohl, & Wetzel, 1994). Surveying angel investors, Haar, Starr, and MacMillan (1989) also address the general market, but they also explore screening criteria, noting key similarities and differences from formal VC investors.

As the domain of venture financing continued its evolution, more theory-driven work emerged—particularly with VC research. The principal-agent framework, for example, emerged as a common way to characterize the investor-entrepreneur relationship (Amit, Glosten, & Muller, 1990; Fiet, 1995; Sapienza & Gupta, 1994; Van Osnabrugge, 1998).

Elitzur and Gavius (2003) modeled moral hazard issues that arise as entrepreneurs, angels, and venture capitalists interact. Further examinations of the contractual arrangements between venture capitalists and entrepreneurs became a fruitful stream of research (e.g., Black & Gilson, 1998; Gompers, 1995; Kaplan & Strömberg, 2003; Sapienza & Gupta, 1994). Related work continued to establish theoretical foundations for understanding the extent to which venture capitalists add value and professionalize portfolio companies beyond that of financial contributions (Hellmann & Puri, 2002; Sapienza, 1992; Sapienza, Manigart, & Vermeir, 1996).

When venture capitalists get involved, there is a certification effect that lessens the cost of going public and increases the net proceeds to the offering firm (Megginson & Weiss, 1991). Complementary to the insights provided on the certification role of venture capitalists and the networking of venture capitalists to invest through syndication (Bygrave, 1987), important work emerged on the interorganizational networks of young companies and venture capitalists (Sørensen & Stuart, 2001). Entrepreneurial ventures often benefited by endorsements from reputable exchange partners as reflected in reaching IPO more quickly and obtaining higher valuations than firms without such endorsements (Stuart, Hoang, & Hybels, 1999). Founders who have relationships that provide access to additional resource endowments and investors, it was quickly understood, are more likely to receive funding (Shane & Stuart, 2002). Of course, some relationships matter more than others, and they likely vary by the types of uncertainty (Gulati & Higgins, 2003). Additional work leveraged theory to open up the decision-making processes of investors, acknowledging the irrational and often biased nature of such decisions (Shepherd, 1999; Zacharakis & Meyer, 1998; Zacharakis & Shepherd, 2001).

Through this overview of some of the early research, it is clear that important foundations were laid. Building from these works, we now address how researchers have pushed forward with this important frontier.

Venture Financing Research: 2004 to 2016

Research from 2004 to 2016 expands upon the work above, diving deeper into VC while channeling greater attention to additional types of funding. This research includes a total of 263 paper classifications. We now turn to VC, CVC, and angel research within this time period, reviewing research at the individual, organizational, and market levels. Furthermore, financing mechanisms are reviewed in this order, paralleling the scholarly impact (indicated by citation count and article volume) of articles falling within each.

VC

Venture capitalists—professional investors funding portfolios of potentially high-growth ventures—have had a transformative impact on the modern entrepreneurial landscape. Thus, this component of the financing landscape has drawn more research attention than any other form of equity financing; a robust body of research has examined topics across the individual, organizational, and market levels, with the greatest attention being paid to the organizational level. In total, VC research represents over three fourths of the papers under consideration.

VC Individual Level

From the venture capitalist's perspective, individual-level research largely focuses on the ways investors evaluate prospective deals. Related research takes the perspective of the entrepreneur and focuses on why some ventures are funded while others are not. A third stream of research focuses on the venture capitalist–entrepreneur dyad and how both parties interrelate.

Venture capitalists. Investigations into how venture capitalists evaluate prospective opportunities have attracted significant attention. Early research in this stream established self-reported criteria that venture capitalists use to make decisions (e.g., management team, market characteristics). In painting a more nuanced picture, research has since progressed to draw attention to the subjective, interactive, and contingent nature of VC evaluations that defy rational formulas and rigid approaches (Kirsch, Goldfarb, & Gera, 2009; Petty & Gruber, 2011). For example, underscoring the importance of individual-level difference considerations, novice venture capitalists tend to focus on the qualifications of individual venture team members, while more experienced venture capitalists emphasize broader team cohesion (Franke, Gruber, Harhoff, & Henkel, 2008). Other research evaluates the perceived motivations of the entrepreneurs—entrepreneurial passion—and the preparedness of the team, indicating that preparedness more positively affects decisions to invest than does passion (Chen, Yao, & Kotha, 2009). Furthermore, Matusik, George, and Heeley (2008) demonstrate that a venture capitalist's values influence evaluations of the management team, while other advances delineate trade-offs between perceived control and the prestige of the management team, suggesting that venture capitalists are more willing to accept less control as the prestige of the management team increases (Drover, Wood, & Payne, 2014). Extant research also highlights VC decision making from a longitudinal vantage point, where the importance of decision criteria varies across evaluative stages (e.g., preliminary screening, due diligence; Petty & Gruber, 2011).

Overall, progress in this area emphasizes that venture capitalists utilize multiple, interacting data sources and commonly rely on inferential logic more than on the planning documents produced by the entrepreneurial team (Kirsch et al., 2009). Promising advances are coming from models probing cognitive processes that capture the subjective, contingent, and multifaceted nature of VC decision making.

Entrepreneurs. VC research has also probed interesting angles involving individual entrepreneurs. Kaplan, Klebanov, and Sørensen (2012) examine the general ability and the execution skills of CEOs in privately funded ventures and find both types of human capital positively relate to subsequent performance. Schwienbacher (2007) categorizes entrepreneurs into three types—lifestyle, serial, and pure profit maximizing—and finds the category of entrepreneur to affect the source of financing pursued. Ueda (2004) considers the characteristics and trade-offs associated with the entrepreneurs' pursuit of venture capitalists versus banks, where factors such as strength of intellectual property (IP) and collateral emerge as important considerations. Ethical perceptions of entrepreneurs have also garnered some attention: Entrepreneurs' perception of a prospective investor's ethical reputation affects their willingness to partner, indicating that entrepreneurs become increasingly tolerant of less ethical venture capitalists as the need for funding becomes more severe (Drover, Wood,

& Fassin, 2014). Moreover, in selecting venture capitalists, it has been shown that entrepreneurs are willing to pay a premium to partner with more reputable venture capitalists by way of a discounted valuation (Hsu, 2004).

The investor–entrepreneur dyad. In addition to the above, the relationship between venture capitalists and the entrepreneurs that they fund holds important implications for venture outcomes. From a funding standpoint, Franke, Gruber, Harhoff, and Henkel (2006) find that venture capitalists tend to look more favorably on teams that have training and professional experience similar to themselves. Bruns, Holland, Shepherd, and Wiklund (2008) find that similarities between the specific and general human capital of the funding agent and the entrepreneurs are both significant indicators of funding. In a similar vein, Bengtsson and Hsu (2015) find that coethnicity between the entrepreneur and VC partner increases the likelihood of VC investments. However, they also find that shared ethnicity results in less desirable financial outcomes. Murnieks, Haynie, Wiltbank, and Harting (2011) examine the way entrepreneurs and venture capitalists “think” as reflected in their decision-making processes and find that venture capitalists tend to favor new investment opportunities led by entrepreneurs who think in ways similar to themselves. Accordingly, it seems that investors are attracted to entrepreneurs who are like them in some important ways. Clearly, the influence of similarity biases is important for understanding how venture capitalists decide whether to invest in specific deals.

VC Organizational Level

VC research at the organizational level is a central area of inquiry in the entrepreneurial finance literature. A significant portion of this focus centers on venture capitalists’ mitigation of risk, the effects of VC firm intermediation on portfolio companies, and the certification role venture capitalists provide.

VC risk mitigation strategies. Given the high potential of goal conflicts between investors and entrepreneurs, coupled with possible adverse selection, venture capitalists regularly utilize risk mitigation strategies to manage their investments (Cumming, 2008; Hellmann, 2006; Kaplan & Strömberg, 2004; Tian, 2011). Much of the literature addressing agency concerns from the perspective of venture capitalists explores various cash flow rights, control rights, and incentives, given that such mechanisms are increasingly used when risks and complexity are higher (Kaplan & Strömberg, 2004).

In volatile environments with asymmetric information, venture capitalists regularly utilize multistage investment vehicles (i.e., providing capital in a series of allocations over time vs. one lump sum; Grenadier & Malenko, 2011; Li, 2008; Tian, 2011). In so doing, venture capitalists can limit their exposure to both agency and developmental risks, deciding whether to continue investing, withdraw, or even renegotiate terms/valuation at various stages (Guler, 2007; Li & Chi, 2013; Tian, 2011). The structure and approach to staging can vary; for example, venture capitalists tend to invest through more rounds in smaller increments when there is greater geographic distance between a VC firm and the investee firm (Tian, 2011). That said, as a result of intraorganizational politics as well as pressure from coinvestors and limited partners (escalating commitment; Guler, 2007), venture capitalists do not always terminate poor performing investments.

Venture capitalists regularly utilize other contractual mechanisms, such as stock options (Arcot, 2014), covenants (Bengtsson, 2011), convertible securities (Hellmann, 2006), board representation (Wijbenga, Postma, & Stratling, 2007), and active monitoring of the management team postinvestment (Yoshikawa, Phan, & Linton, 2004). Recent work on security design in VC contracts emphasizes the need for more context-specific research into the ways various types of securities are utilized by investors (Burchardt, Hommel, Kamuriwo, & Billitteri, 2016). While it is suggested that differences in tax regimes and institutional practices explain some of these differences, related research also predicts that the use of convertible preferred equity can reduce information asymmetries about entrepreneurial abilities and better align entrepreneur–investor interests (Arcot, 2014). When fixed payoffs are higher, investors often utilize contractual covenants to try to control behaviors (Bengtsson, 2011); still, the enforceability of these contracts is contingent upon either the investor’s reprisal abilities (Lu, Hwang, & Wang, 2006) or the institutional environment in which the investor is operating (Li & Zahra, 2012). Overall, it is often recognized that VC control tends to decrease as the investee company performance improves over time.

Venture capitalists also use syndication as a way to mitigate risk (Manigart et al., 2006). Here, multiple venture capitalists invest alongside one another to form a syndicate (Gu & Lu, 2014; Keil, Maula, & Wilson, 2010). Through syndication, access to investment opportunities are shared and often jointly vetted (Hochberg, Ljungqvist, & Lu, 2007); thus, venture capitalists can spread their investment risks across a larger pool of promising deals as opposed to investing larger amounts into a smaller number. Syndication has well-documented performance consequences—particularly with regard to one’s network position (Hochberg et al., 2007; Sørensen & Stuart, 2008).

VC intermediation. A robust stream of work investigates important questions regarding the role of venture capitalists in selecting and adding value to portfolio companies (e.g., strategic guidance) and exploring important contingencies therein (Chemmanur, Krishnan, & Nandy, 2011; Dimov & Shepherd, 2005; Sørensen, 2007). It is widely recognized that VC investments are inherently a two-sided matching process: High-quality venture capitalists search for high-quality portfolio companies (Sørensen, 2007). The impact and outcomes of VC involvement on portfolio companies, though, varies.

The effectiveness of VC intermediation activities on new ventures appears to vary on the basis of how outcomes are measured. Fitza, Matusik, and Mosakowski (2009) estimate that venture capitalists contribute about 11% to variance in interround *valuation* changes of the portfolio companies in which they invest. They also find no evidence of venture capitalists’ ability to select ventures with higher potential valuation, but they do note that the influence of VC intermediation activities on firm valuation is more pronounced during the early investment period. Related research reports that venture capitalists more likely select firms with higher growth potential and that VC intermediation activities increase the growth of factor productivity in firms by 18% (e.g., revenue growth; Chemmanur et al., 2011; Croce, Martí, & Murtinu, 2013). Importantly, this research also estimates that even firms that were not originally backed by venture capitalists would have experienced approximately 5% increase in factor productivity growth had venture capitalists invested in the ventures (Chemmanur et al., 2011). At the same time, other research reports that VC-backed firms achieve higher

levels of revenue growth but not necessarily higher levels of profitability (Puri & Zarutskie, 2012). Puri and Zarutskie (2012) observe that VC-backed ventures experience more rapid growth and are more likely to survive for 5 years as opposed to non-VC-backed firms, but beyond 5 years, this survival likelihood appears to reverse.

Extant research also zooms in on contingent relationships, where a meta-analysis of 76 samples (36,567 VC-backed firms) suggests that positive performance effects largely hinge on situational factors such as stage of development and industry (Rosenbusch, Brinckmann, & Müller, 2013). Another contingency accounting for variation in performance is heterogeneity across VC firms (Dimov & Shepherd, 2005; Kaplan & Schoar, 2005; Sørensen, 2007). The abilities of venture capitalists to evaluate potential investment targets and add value are not always equal; Sørensen (2007) reveals that VC firms with greater experience are more likely to realize IPO exits than their lesser-experienced counterparts—through selection *and* influence (though to differing degrees). Adopting a human capital approach, Dimov and Shepherd (2005) find that VC firms composed of higher levels of general human capital had a positive impact on portfolio company IPOs, while surprisingly the specific human capital of venture capitalists did not affect venture outcomes.

Other work establishes that both high- and low-reputation VC firms influence portfolio firm efficiency but cast differential effects (Chemmanur et al., 2011). Moreover, scholars have focused on contingencies associated with the investee firm: The value venture capitalists provide may be greater among lower-quality ventures (Baum & Silverman, 2004). Although team makeup and capabilities play a key role in attracting VC funding (Beckman, Burton, & O'Reilly, 2007), the value of venture capitalists may be greater for firms with incomplete teams. The number of investee companies (portfolio size) is also likely to influence the constructive coaching and value provided by venture capitalists given the division of time, attention, and resources (Fulghieri & Sevilir, 2009).

Other studies point to a less-positive or mixed view of the value that venture capitalists bring (Baum & Silverman, 2004; Busenitz, Fiet, & Moesel, 2004); Busenitz and colleagues (2004), for example, tracked the performance of VC-backed companies over a 10-year period and report that the strategic guidance provided by venture capitalists plays no role in increasing the odds of a successful exit. Taken together, significant attention has been paid to disentangling the relative value provided by venture capitalists.

Certification. Although much research revolves around the VC impact and value-added services on portfolio firm performance, other studies explore the certification value of VC ties on portfolio company performance—especially during exit events like IPOs (Pollock, Chen, Jackson, & Hambrick, 2010). Backing by higher-reputation venture capitalists is related to more favorable exits, and the share prices of these companies is frequently higher than non-VC-backed companies (Nahata, 2008). The certification effects of VC ties also appear to enhance the formation of alliance networks both among VC firms and among portfolio companies (Gu & Lu, 2014; Hoehn-Weiss & Karim, 2014; Lindsey, 2008). Additional research suggests these firms are subject to underpricing as a result of VC grandstanding (P. M. Lee & Wahal, 2004), which may positively affect future VC deal flow but generally reduces the gains to the original owners. Clearly, VC involvement can communicate important social signals, but the complexities of such activity are still being teased out.

VC Market Level

VC market-level research falls broadly into two categories. One group focuses on exogenous (market) factors that shape VC organizational-level decisions and outcomes, and so is related to the themes raised in the prior section. Another examines country-level outcomes associated with VC.

Exogenous (market) factors and organizational-level action. VC investment decisions are shaped by where others are investing (Hochberg, Ljungqvist, & Lu, 2010; Inderst & Müller, 2004; Nanda & Rhodes-Kropf, 2013; Sørensen & Stuart, 2008), as well as the presence of government funding, though in ways that vary by country (e.g., Guerini & Quas, 2016; Munari & Toschi, 2015). Additionally, the development of the legal environment affects deal screening and syndication behavior (Cumming, Schmidt, & Walz, 2010). The institutional context also affects syndicate patterns (Gu & Lu, 2014) and the amounts and timing of investments (Dai, Jo, & Kassicieh, 2012), as well as postinvestment governance decisions (e.g., Cumming et al., 2010).

In terms of outcomes, cultural and institutional distance can negatively affect successful exits (Li, Vertinsky, & Li, 2014), and network ties can ameliorate negative effects associated with cultural and geographic distance (Jääskeläinen & Maula, 2014). Other work (in Asia) indicates that investments that have both foreign and local venture capitalists are more likely to have successful exits than those with only domestic or foreign firms (Dai et al., 2012), while the type of equity investment and associated levels of retained equity can affect IPO performance in different institutional contexts. Bruton, Filatotchev, Chahine, and Wright (2010) find that VC-retained equity in the United Kingdom context is positively related to IPO valuations, while angel-retained equity is positively associated to IPO valuations in France, suggesting the legal traditions of these two countries account for such differences.

Overall, the level of activity in a market and formal institutions conducive to markets generally have positive effects on VC entry, involvement, and outcomes (Cumming et al., 2010; Nanda & Rhodes-Kropf, 2013; Sørensen & Stuart, 2008), while cultural and institutional distance are generally negatively related to successful outcomes (e.g., Li et al., 2014). Foreignness and relative level of institutional development in a venture capitalist's home country may affect VC decisions on where and how to invest (Gu & Lu, 2014), and different forms of risk capital may be optimal in different institutional contexts (Bruton et al., 2010). Additionally, syndication between home and host country VC firms generally yields positive outcomes, especially in contexts with weaker formal institutions (e.g., Jääskeläinen & Maula, 2014).

Country-level outcomes. Government incentives, formal institutions, and informal institutions can shape VC investment patterns at the country level of analysis. Extant work on incentives has examined government initiatives aimed at providing VC. For example, Cumming and MacIntosh (2006) find evidence that a Canadian government venture vehicle crowds out private venture money. On the other hand, the government-run Innovation Investment Fund program in Australia, which is somewhat unusual relative to other government incentives in that it focuses on government-private sector investments, has largely been successful (Cumming, 2007). Overall, results from seeding the VC market with government funds are mixed.

Recent research on formal institutions finds that institutions that enable economic integration in the form of a common market and currency positively relate to cross-border VC flows

in these countries (Alhorri, Moore, & Payne, 2008) and that formal institutions as represented in the World Governance Index (e.g., rule of law, IP protection, political stability, accountability, control over corruption) have a positive effect on country-level volume of VC investment (Li & Zahra, 2012). Furthermore, developed VC markets may attenuate the negative effect that stringent bankruptcy laws have on entrepreneurship levels (S. H. Lee, Peng, & Barney, 2007).

The reality that strong formal institutions can be lacking, especially in emerging economies, has led to work that highlights how networks can act as substitutes for strong formal institutions in some country contexts, encouraging VC activity (Ahlstrom & Bruton, 2006). Furthermore, patterns of migration between countries can affect cross-border investment levels (Iriyama, Li, & Madhavan, 2010), and the nature of social networks can shape funding patterns of early-stage ventures and governance of those ventures (Bruton, Fried, & Manigart, 2005), reinforcing the role that informal social relationships can play in driving VC investment across borders.

Moreover, informal institutions in the form of values and norms affect country-level VC patterns. Uncertainty avoidance, for example, moderates the role of formal institutions on country levels of VC investment (Li & Zahra, 2012), and the value placed on entrepreneurship can shape funding patterns of early-stage ventures and their governance (Bruton et al., 2005).

The findings at the country level indicate that government incentives yield mixed results, while market-based formal institutions and personal connections positively affect VC investment levels. Furthermore, theoretical work suggests that while many macrolevel efforts have been aimed at developing risk capital markets as a means of generating robust entrepreneurial ecosystems, such efforts are insufficient if they are not also accompanied by contributory factors such as novel ideas, role models, leadership, and access to large markets (Venkataraman, 2004). Though this theoretical work conceptualizes VC market activities as a separate factor in entrepreneurial ecosystem development, the legitimacy and value placed on entrepreneurship may actually be a necessary condition for the development of VC markets themselves (e.g., Bruton et al., 2005).

CVC

The impact of CVC investing—where existing corporations seek minority equity stakes in young ventures—has triggered a growing body of research. Scholars have investigated a number of important topics in this emerging space, predominately at the organizational level. Across venture financing papers identified as part of our article set, CVC accounts for around 10%.

CVC Individual Level

Only a handful of CVC studies exist at the individual level. The lacuna of work is partially due to limited data and partially due to the focus on organization-level dynamics prevalent in the corporate setting. Dokko and Gaba (2012) as well as Hill and Birkinshaw (2014) discuss CVC personnel and their career backgrounds while both Souitaris and Zerbinati (2014) and Souitaris, Zerbinati, and Liu (2012) bring qualitative insights into corporate venture capitalists' mind-sets and investment rational. Moreover, Dushnitsky and Shapira (2010) document individual compensation schemes and their impact on subsequent investment practices.

CVC Organizational Level

Research at the organizational level falls into four main categories: studies examining antecedents to CVC decisions, those that examine the structure of the CVC unit itself, those that look at CVC outcomes, and, finally, studies that examine CVC from the perspective of the firm receiving the investment.

Antecedents. The decision to engage in VC investment is affected by both economic and behavioral considerations. Dushnitsky and Lenox (2005a) view CVC investment as part of a firm's optimal innovation strategy in which firms weigh the marginal innovative output of CVC activity against that of internal R&D. They find that industry-level factors (i.e., technology ferment and patenting activity, role of complementary assets, and IP regime) as well as firm-level resources (i.e., absorptive capacity, cash flow availability) stimulate CVC activity. Additional work confirms the role of the aforementioned industry and firm-level factors and points to a subtle interaction among the two (Basu, Phelps, & Kotha, 2011); while industry technological ferment and resource-rich firms are two factors associated with higher CVC activity, resource-rich firms are less—rather than more—likely to pursue CVC in industries that experience ferment. Relatedly, a firm is more likely to undertake CVC, rather than merger and acquisition (M&A) activity, in the face of high exogenous uncertainty (i.e., when market uncertainty is high; Tong & Li, 2011).

There are also behavioral drivers of these venturing activities. CVC practices have diffused via contagion from VC firms and then within an industry population (Gaba & Meyer, 2008). Additionally, when innovation performance is above aspiration levels, a firm is less likely to initiate or continue existing CVC activities and is also less likely to launch a CVC unit when performance is below aspirations (Gaba & Bhattacharya, 2012).

CVC unit. CVC activity is situated within a part of the corporate organization. Extant work studies four facets of this unit's activity: objectives, structure, staffing, and salary. The objective of corporate investors received significant attention in early studies. Some firms pursue CVC to secure financial gains in a manner similar to independent VC funds, while many CVC units pursue strategic benefits for their parent firm. These may take the form of exposure to novel technologies, access to complementary products and services, entry to new markets and geographies, and so on (Chesbrough, 2002; Keil, 2004; Keil, Autio, & George, 2008).

CVC also varies by the structure of the unit (Dushnitsky & Shaver, 2009; Hill, Maula, Birkinshaw, & Murray, 2009), the personnel it employs (Dokko & Gaba, 2012; Souitaris et al., 2012; Souitaris & Zerbini, 2014), and incentives offered (Dushnitsky & Shapira, 2010; Hill et al., 2009). First, there is substantial variation in CVC structures (Hill et al., 2009). Some CVC units follow VC funds. They are structurally separated from the parent corporation, manage a dedicated pool of capital, and have full investment discretion. Others are embedded within a business unit and request approval and funding on a deal-by-deal basis. The former structure is associated with greater success in realizing financial objectives, whereas the latter is associated with strategic gains.

Second, human resource strategies vary. As a corporate in-house VC arm, some units are staffed by long-term corporate employees. These units commonly pursue fit with internal stakeholders (e.g., business units; Souitaris et al., 2012; Souitaris & Zerbini, 2014), and their portfolio companies are likely to be acquired and assimilated by the parent firm (Dokko

& Gaba, 2012). Other units hire career venture capitalists. Accordingly, they seek legitimacy with external stakeholders (i.e., independent venture capitalists), and their portfolio companies are more likely to achieve financial returns. Taken together, the past career of CVC personnel shapes their views and practices.

Third, incentive schemes are important (Dushnitsky & Shapira, 2010). Venture capitalists often share in the financial success of their portfolio companies. They can expect windfalls in the millions of dollars through carried interest; however, pressures to maintain pay equality across the corporation imply that many corporate venture capitalists receive little more than straight salary. Consequently, incentive schemes are often adopted with little regard to CVC objectives or staffing. This can lead to conservative investment practices (Dushnitsky & Shapira, 2010) and undermines the performance of the unit as a whole (Benson & Ziedonis, 2010; Dushnitsky & Shaver, 2009; Hill et al., 2009).

Outcomes of CVC investment. CVC investments interact with a firm's alliance and acquisition activities and can affect its innovation outcomes and financial performance. Strategic alliances and joint ventures—as well as CVC—yield the greatest innovation contribution when targeting partners in related industries (Keil, Maula, Schildt, & Zahra, 2008). Moreover, a comparison of CVC-investing firms to noninvesting peers reveals that the former produces higher rates of innovation (Dushnitsky & Lenox, 2005a). Along these lines, evidence from the medical device industry illustrates that parent firms' products incorporate innovations from their portfolio companies (Smith & Shah, 2013). Also, the contribution of CVC activity to patenting rates increases linearly with the depth of involvement of the corporate venture capitalist with its portfolio companies (Wadhwa & Kotha, 2006) and exhibits an inverted U-shaped relationship to the overall diversity of the portfolio (Wadhwa, Phelps, & Kotha, 2016). CVC relationships are also associated with timely attention to emerging technological discontinuities (Maula, Keil, & Zahra, 2013).

Hill and Birkinshaw (2014) found success to be dependent on a unit's ability to build strong relationships internally (i.e., with senior executives as well as business unit managers) and externally (i.e., with independent VC funds). There are also direct relationships between VC firms and CVC firms. These relationships facilitate access to favorable investment opportunities and afford learning of investment practices (Hill et al., 2009; Souitaris & Zerbini, 2014), but the resources of CVC firms can substitute for their lack of prior centrality and allow them rapidly to gain central positions in VC syndicates (Keil et al., 2010).

Furthermore, CVC investments can affect strategic alliance formation. For example, an inverted U-shaped relationship exists between a firm's CVC investments and its alliance formation (Dushnitsky & Lavie, 2010). In addition, a prior CVC relationship between a corporation and a start-up increases the likelihood the two will subsequently enter a strategic alliance (Van de Vrande & Vanhaverbeke, 2013). Mathews (2006) advances an alternative view and predicts that a CVC investment will follow, rather than lead to, a strategic alliance.

CVC also affects acquisition activity. Interestingly, Benson and Ziedonis (2010) find that on the basis of acquisition premia, acquirers can be worse off when acquiring start-ups they have previously funded. They also find a positive contribution to the firm's overall M&A strategy. Specifically, they find a positive CVC-M&A association that is strengthened for firms with (a) a dedicated CVC unit or (b) an intense CVC budget relative to internal R&D.

Finally, CVC may have an impact on financial performance. The performance of the CVC unit varies with a large number of units experiencing substantial negative returns and only

some delivering top-tier returns (Allen & Hevert, 2007). Firms that engage in CVC for strategic objectives contribute more to overall parent firm financial performance. In contrast, those that pursue VC-like objectives can erode the parent's performance (Dushnitsky & Lenox, 2006). Additionally, there is evidence of a U-shaped relationship between CVC portfolio diversity and corporate financial performance (Yang, Narayanan, & De Carolis, 2014).

Start-up's perspective and performance. To understand fully the CVC phenomenon, one has to consider the start-up's perspective. Dushnitsky and Shaver (2009) note that the decision to undertake an investment from an industry incumbent is not a trivial issue. On one hand, a corporate investor may offer multiple benefits. On the other hand, the corporation may imitate the invention. As a result, entrepreneurs may forego corporate venture capitalists altogether and pursue funding from a venture capitalist. Indeed, Dushnitsky and Shaver find that CVC investment is less likely in the face of heightened imitation concerns, namely, when (a) the two are prospective competitors in the same industry and (b) the industry IP regime is weak. A survey of start-up CEOs further corroborates these dynamics (Maula, Autio, & Murray, 2009). A start-up is more likely to interact with its corporate investor, and less likely to adopt relationship safeguards, when the pair pursues complementary (rather than potentially competing) offerings.

There is also some evidence as to the impact on start-up's ultimate performance. CVC-funded start-ups perform at least as well as their VC-backed peers. The positive CVC effect is strongest when the start-up is well positioned to take advantage of corporate complementary assets. When that is the case, CVC-backed start-ups are more likely to go public (Park & Steensma, 2012), less likely to experience underpricing at IPO (Wang & Wan, 2013), and more likely to exhibit a higher rate of innovation output (Chemmanur, Loutskina, & Tian, 2014; Garrido & Dushnitsky, 2016).

CVC Market Level

CVC research at the market level focuses on the trends and broader outcomes of CVC investment activity. Research suggests that over the past 50 years, CVC exhibits highly cyclical patterns, both in terms of total investment amounts and in terms of the number of firms that engage in CVC. Studying CVC at the aggregate level, scholars have found, for example, that an increase in the total R&D expenditures within an industry is associated with a greater number of CVC investments in that industry (Sahaym, Steensma, & Barden, 2010). Notably, CVC is increasingly becoming a global phenomenon with many impactful CVC investors outside the United States or Europe.

Next, and equally importantly, the adoption of CVC practices is part of a broader transition in corporate R&D strategies, shifting away from an exclusively internal effort and towards embracing external sources of innovations (also known as open innovation). Put differently, many established firms pursue CVC not merely as a way to generate high financial returns but, rather, as a critical vehicle to engage and nurture relationships with an external community: that of innovative entrepreneurial ventures. Indeed, there is evidence that a firm's CVC activity is associated with its allocation of resources towards innovation input (i.e., R&D spending) as well as its innovation outputs (i.e., patenting output; Dushnitsky and Lenox, 2005b). Because focus is strategic rather than financial, it is therefore not surprising

that a notable number of CVC units have persisted across several cycles of boom and bust in the VC industry.

Institutions and regulations that affect innovation-oriented entrepreneurship play a role in driving Fortune Global 500 to adopt CVC (Da Gbadji, Gailly, & Schwienbacher, 2015). Companies in countries with a developed market for early-stage investments are more likely to engage in CVC, while costly personal bankruptcy regulations are associated with lower uptake of CVC practices. Interestingly, local conditions have little impact on the degree of internationalization (i.e., investment in start-ups based in foreign innovation hubs). Also, as noted earlier, proliferation and success of VC funds is an important factor in firms' adoption of CVC (Gaba & Meyer, 2008), and start-ups may forego CVC and opt for VC in institutional contexts when imitation concerns are heightened (Dushnitsky & Shaver, 2009; Maula et al., 2009).

Angel Investment

Angel investors—individuals investing their own capital independently or through angel groups—are playing an increasingly important role in funding young, high-growth-potential ventures offering some of the earliest stages of funding. While the influence of angel investment is substantial, it has attracted only minimal research attention (Huang & Pearce, 2015; Wiltbank, Read, Dew, & Sarasvathy, 2009). Studies investigating angel investment represent just under 10% of the articles in our paper set. Kerr et al. argued, “Angel investors have received much less attention than venture capitalists, despite the fact that some estimates suggest that these investors are as important for high-potential startup investments as are venture capital firms” (2014: 21).

Angel Individual Level

Angel research has taken an interest in individual investor decisions, in particular the internal processes and approaches of angel investment decision making (Wiltbank et al., 2009). Maxwell, Jeffrey, and Lévesque (2011) found that angels utilize cognitive processes that tend to be stage dependent: First, heuristic shortcuts are employed to pare down the large number of prospective opportunities (elimination by aspects), and a more in-depth and systematic evaluative approach is then utilized to assess a smaller subset of ventures. Huang and Pearce (2015) argued that angels rely on intuition, heuristic-based reasoning, and that intuitive evaluations more accurately led to selecting higher-success investments. Investigations into the role of individual-level investor characteristics have begun to show that these characteristics influence the interpretation of evaluative criteria (Mitteness, Sudek, & Cardon, 2012). Additional research links the angel's approach to investment outcomes, finding that an investor emphasizing control logic (i.e., attempting to control the future) versus prediction logic (i.e., attempting to predict the future) in evaluations negatively correlates with an angel's investment losses (Wiltbank et al., 2009). These studies together advance the nexus of investment opportunities under evaluation and the corresponding cognitive processes that compose the investor's assessment.

The entrepreneurs in the angel funding context have received limited attention. Inquiry here has sought to understand both economic and behavioral motives that drive the entrepreneurs' pursuit of angel capital. Fairchild (2011) explores the trade-offs that entrepreneurs face between

pursuing angel and VC funding, considering the consequences therein. Other research takes a dyadic approach, simultaneously exploring both individual angel and entrepreneur interactions. Such work establishes that conflict influences each party's intention to exit (Collewaert, 2012) and the innovation levels that result (Collewaert & Sapienza, 2014). Bammens and Collewaert (2014) explore intrateam trust perceptions between individual angels and lead entrepreneurs, establishing that such perceptions play a role in shaping performance evaluations.

Angel Organizational Level

Angel research at the organizational level examines angel investment organizations (i.e., angel groups) as well as the entrepreneurial organizations that receive angel funding. Angel group research, while emerging only recently, has begun exploring group decision criteria. Carpentier and Suret (2015) articulate some differences between the processes and criteria utilized by angel groups and individual angels, finding angel groups tend to focus more on risks associated with market/execution than on agency risks and are more concerned with exit. Other research investigates the gender composition of angel groups, delineating ways in which variations in group composition influence investment likelihood (Becker-Blease & Sohl, 2011). Angel group research also examines the impact of angel group involvement on investee organizations. This research indicates that funding from certain angel groups can influence venture performance. For example, Kerr et al. (2014) found that organizations that sought and received angel group funding (vs. ventures that sought funding but were narrowly rejected) tend to exhibit superior performance. In a related vein, Dutta and Folta (2016) explored how an organization's innovation activity and exit is affected differently by angel group versus VC involvement, offering a first comparison of effects resulting from angel group involvement and other established players. They found, for example, that VC-backed ventures produce higher levels of innovation and commercialize faster.

Relatedly, considering the investee organization, Bruton et al. (2010) demonstrate that angel investors provide "value-enhancing effects" that are manifest in IPO performance (Bruton, Chahine, & Filatotchev, 2009). Regarding organizational slack resources, angel involvement can positively affect the relationship between certain slack resources and performance (Vanacker, Collewaert, & Paeleman, 2013).

Angel Market Level

Angel research at the market level compares and contrasts angel markets with other mechanisms, such as VC markets, in an effort to understand how these markets differ as well as interact with one another. Hellmann and Thiele (2015), for example, argue that the angel and VC markets are "friends," where angels provide venture capitalists with a source of deal flow, while venture capitalists provide needed follow-on funding. Still, these markets are simultaneously "foes" in that angel and VC collaborations often clash around valuations, value add, and so forth. Other research examines how broader social factors affect angel markets. Here, research investigates gender differences in the supply of angel capital, noting the existence of systematic differences in how male and female angels both allocate funds and network with others within the marketplace (Harrison & Mason, 2007). Ding, Au, and Chiang (2015) examine country-level differences in social trust, finding that countries with higher levels of social trust reflect heightened levels of angel investment.

Future Research

The dynamic nature of venture financing continues to challenge existing paradigms and create new research opportunities. Following the above review outline, we now discuss promising research opportunities, starting with VC.

Directions for Future VC Research

Although VC research is the most mature segment of research in the equity financing literature, important questions remain. The inherent uncertainty of VC decision making has led multiple researchers to probe VC decision processes and evaluation criteria. While important advancements have been made in this area, much of this progress focuses on decision processes of individual venture capitalists or individual VC firms. However, in practice, venture capitalists generally share risk by syndicating investments, and VC decision making tends to be a collective effort. Thus, a more socialized perspective *both within and across VC firms* is merited.

Within firms, how are existing assumptions about individual VC decision making challenged or changed when viewing the process as a collection of interacting decision makers? Funds commonly utilize investment committees to make final decisions regarding potential investments. Decision by committee can help decision makers offset the negative influences of individual biases. However, just as decisions to terminate an investment are influenced by political dynamics (Guler, 2007), social influences may also affect initial funding decisions and subsequent interactions with portfolio companies. This leaves open questions about how venture capitalists vary in their group decision-making processes, internal social dynamics, and behavioral risk taking and how this shapes the identification and development of portfolio companies. Across partnering VC firms, how do syndicate dynamics influence VC investment decisions? At a minimum, social dynamics within syndicates can affect follow-on funding, and the prevalence of investment syndicates suggests social dynamics across firms deserve much greater attention.

Future research needs to move beyond the undersocialized perspective represented in many existing studies. Drawing on perspectives involving team mental models/schemas (e.g., Mohammed, Ferzandi, & Hamilton, 2010), group decision processes (e.g., Barsade, 2002), social cues (Banerjee, 1992), or more theory on teams, groups, and social dynamics is needed for a better understanding of VC decision making. The VC setting can also be useful in testing boundary conditions associated with these team, group, and social dynamics theories, many of which have largely been examined within a firm's boundaries; in the VC context, these interactions occur both within and across firm boundaries.

Relatedly, despite the central role of power dynamics established in other relational dyads, few articles have explored the role of power in shaping VC investment activities. Although research has explored how the bargaining power of venture capitalists influences both their relationship with firms and firm valuation, venture financing research is just starting to explore how power and political influences shape the functioning and structure of VC syndicates (Ma, Rhee, & Yang, 2013). Power dynamics likely influence access to VC syndicates, the allocation of shares across syndicate partners, and how lead venture capitalists and other parties sanction or reward syndicate partners. The role of power dynamics is also likely to increase in importance as VC investing becomes more international and crosses institutional

and cultural boundaries (Batjargal, 2007). In addition, though social theory tends to define power negatively, the use of power by investors and entrepreneurs does not always generate conflict but, instead, can produce favorable outcomes for both parties (Hallen, Katila, & Rosenberger, 2014) and can be utilized in a variety of ways (Fleming & Spicer, 2014), such as in providing social defenses for nascent firms. Thus, a more inclusive approach to future interorganizational power research that extends beyond resource- or power-dependence approaches, and explores both the negative and positive uses of power (e.g., soft power; Santos & Eisenhardt, 2009), are important topics.

The widespread use of syndication networks suggests multiple agency issues that venture capitalists face as well as the potential for knowledge transfer among the companies *within* a venture capitalist's portfolio and/or the portfolio of syndicate partners. The implications of these issues for follow-on funding, ways the venture capitalist interacts with portfolio companies that are increasing or decreasing in value relative to other investments, and opportunities for the portfolio company to obtain new sources of knowledge and funding are important topics that deserve further consideration.

Next, because venture capitalists are firms composed primarily of knowledge assets, they present a rich context in which to examine boundary conditions of existing strategy theory that has been developed based on firms with a portfolio of knowledge assets as well as tangible assets. While several studies have used this context to reevaluate the corporate effects (e.g., Fitza et al., 2009) and diversification (Matusik & Fitza, 2012) literature and findings, many opportunities remain for using this context better to refine strategy theory.

One major trend within the VC arena we discussed is the increasing internationalization of VC investment activity. While both formal and informal institutional environments influence the VC activities (Gu & Lu, 2014), much more research is needed to disentangle the influences of formal and informal institutional factors on VC investment activities, particularly in developing contexts. Much of this work largely assumes there is an optimal configuration of formal and informal institutions conducive to VC investment; Bruton et al. (2010) indicate that this is not the case. Different institutional contexts may require different types of investors and investor behaviors. Instead of focusing on how closely the institutions in different countries align with those in the dominant VC market (i.e., the United States), there are opportunities to identify alternative configurations for risk capital markets that may fit with different institutional heritages.

Additionally, complex institutional environments create an interesting set of opportunities and challenges that have not been widely explored. Unique formal and informal institutional arrangements around the world increase the complexity of managing global VC activities. Additionally, venture capitalists are carriers of institutional logics (cf. Almandoz, 2014), so VC actions can shape how entrepreneurial ecosystems evolve and develop with consequential implications for country-level economic growth. Along these same lines, often the category-breaking outlier firms funded by venture capitalists play a role in disrupting and creating novel institutional arrangements. How venture capitalists engage in institutional work to complement these efforts to create, sustain, and disrupt institutional arrangements remains an open question. Certain venture capitalists are public figures and actively utilize their visibility not just to promote portfolio companies but also to challenge the prevailing institutional consensus openly and envision alternative institutional arrangements. These activities are worthy of further attention.

Directions for Future CVC Research

CVC research has the potential to advance our understanding of key issues in entrepreneurship and strategy. A notable feature of CVC—versus other players in the market for entrepreneurial finance—is that it is part of very large (and often global) corporations. As such, the study of CVC brings together theories across the fields of strategy, entrepreneurship, and finance. This affords unique insights into the microfoundations of strategy and the management of innovation and technology as well as corporate strategy and governance.

To date, much of the CVC work has focused on answering the narrow question of the impact of CVC investing of firm patenting output (often, with some contingencies around CVC existing portfolio). Going forward, CVC research may generate deeper insights into the antecedents of CVC as well as the consequences to the parent firm and other key stakeholders (e.g., start-ups and independent venture capitalists). First, CVC work can also draw on—and contribute to—theories on corporate strategy and governance. Why do large firms choose to invest in external start-ups directly rather than empower their shareholders to do so independently? How and why should they organize their corporate venturing activities? More research is needed to explore why structural differences exist across CVC firms and funds. Whereas independent VC funds are largely homogeneous in structure and objective, that is not the case with corporate venture capitalists. Future work should shed light on both the rationale behind and the consequences of these differential structures.

Second, and along these lines, there is room to expand our conceptualization of corporate venture capitalists' overall strategic objective. Whereas exposure to novel technology is an important strategic objective, it is not the only one, and firms likely engage CVC investments to accomplish a broad range of strategic objectives. More research is needed to explore these different objectives and to understand both when corporate venture capitalists choose to invest in other firms and when they choose to terminate these investments. Doing so can offer significant insights into the emerging management of technology and innovation.

Third, the practice of CVC is no longer the prerogative of a handful of West Coast-based corporations. Today, corporate investors are active in almost every continent, including areas that have not traditionally been considered as innovation hubs. As CVC becomes more international, the same challenges we highlight above stemming from institutional complexity will undoubtedly influence these outcomes.

Finally, management scholars have been increasingly exploring the microfoundations of strategy and competitive advantage (Felin, Foss, & Ployhart, 2015). At the heart of the microfoundations conversations are debates regarding the role of human resources in strategic management processes (Felin et al., 2015). CVC provides an intriguing setting to explore these questions since CVC phenomena extend firm-level innovation strategies beyond the boundaries of a firm to source new ideas through investment in other organizations. CVC activities also influence the nature and scope of a firm's alliance strategy. At the same time, emerging evidence suggests that the background characteristics (Dokko & Gaba, 2012; Hill & Birkinshaw, 2014) and mind-sets/investment rationales of corporate venture capitalists influence these activities. Thus, in a sense, CVC practices expand the scope of the firm to engage in knowledge-sourcing strategies that expand the firm beyond its traditional boundaries, and individual-level variables likely play a significant role in these decisions. More research is needed to explore how these firm boundaries decisions are enacted, and individual-level variables will clearly be important to these studies.

Directions for Future Angel Research

Despite an impact that closely rivals or even exceeds formal VC and CVC in the field, research on angel investing remains comparatively sparse (Huang & Pearce, 2015; Kerr et al., 2014). It follows that many opportunities exist for scholars to bolster our understanding of this lesser-studied area of venture financing. Doing so, it is first important to highlight a point of distinction that often goes overlooked in existing research. Angel investors are generally clustered together, but in practice, different types of angel investment exist. These differences bring with them important theoretical and empirical considerations. For instance, independent angels invest on their own, but they can vary from being a novice to fairly sophisticated. Some angels make only a handful of investments, while others are much more advanced (“super” or “serial” angels) and make hundreds of investments. Moreover, angel groups are growing rapidly, introducing yet another dimension for possible inquiry. Thus, we see that the motivations for investing, approaches to investment decisions, managing portfolios, and potential for value add are just a few areas where inherent differences are likely to manifest across angel investors. Key to productive and meaningful advancements is consistent recognition and appreciation of inherent differences across sources of angel funding.

Bearing this in mind, several opportunities exist within this dynamic area of investment. Research on independent angel investment decisions continues to identify how angel investors make decisions and the criteria with which they evaluate potential deals. What makes this area so interesting is that extant research indicates that given the paucity of concrete data to rely on at the venture’s earliest stages, angels are more likely to use intuitive decision-making processes (Huang & Pearce, 2015). Heuristic-based reasoning is most observable and perhaps most effective when embedded within executive control (cognitive) systems that are largely formed through prior experiences (Mitchell, Friga, & Mitchell, 2005). Additionally, angels have multiple motives and broad experiences that influence their decision making. More explicit attention to these issues and how they affect decision making is warranted.

Additionally, *angel groups* are proliferating across the globe. Looking at investment decisions within angel groups is likely to involve additional paradigms. Multiple investors are often making evaluations in concert, bringing a host of critical social structures and dynamics absent in independent angel evaluations. As noted with VC decision making, a more socialized approach may also have relevance for angel groups. How interactions across group members, or a collective approach, shape angel group investment decisions is critical to advancing decision making in this emerging space. Moreover, structural and strategic differences among angel groups are important to consider. Finally, researchers could begin to model variables that account for variance in group performance, including virtual versus in-person groups, group-level measures of members (collective experience or prestige), geographic proximity, due diligence, and syndication patterns. Linking different configurations of angel group characteristics, behaviors, and processes to longitudinal return performance remains a critically important line of inquiry.

The dearth of angel research at the market level also provides many opportunities to examine the implications of recent changes in the angel landscape. For instance, the supply of in-person angel financing has tended to be geographically concentrated (e.g., 75%–80% of angel group investments are within their home region; Halo Report, 2015), but the rapid year-over-year proliferation of in-person groups in new areas brings funding to locations that have been traditionally lacking. Also, the growth in online angel platforms challenges long-held

geographic boundaries of in-person investing, and deals are receiving angel funding that cross borders in new ways. Together, the evolving dynamics of in-person and online angel funding are profoundly shifting access to such capital. Research that probes the institutional forces and social networks (syndication) that govern these evolving funding sources should be beneficial. Market data are also needed that capture performance across different types of angel investment activity.

Directions for Emerging Financing Mechanisms

VC has historically dominated venture finance research, despite venture capitalists' funding less than one quarter of 1% of all new ventures (Kaplan & Lerner, in press). As new forms of entrepreneurial financing, such as equity crowdfunding and accelerators, are emerging, greater attention to other funding sources is important for a fuller understanding of high-growth-potential financing. While such inquiry is likely to become vital, we urge scholars to build from what we have already learned from 30-plus years of venture financing research. As our review makes clear, a rich tradition of research has generated key insights around the funding process. Existing theories and approaches, such as agency theory, information economics, network theory, social dynamics, and cognition, can aid in informing emerging funding sources. More explicitly, how do emerging sources complement or challenge existing theories and assumptions, such as traditionally held geographic assumptions and value-add dynamics? Advances are sure to be accelerated when studying emerging sources within the larger landscape of existing venture financing research.

Equity crowdfunding. With regard to crowdfunding, our focus here is on the equity side. Emerging platforms are beginning to have a significant impact, thus creating important research opportunities (Agrawal, Catalini, & Goldfarb, 2016; Ahlers et al., 2015; Allison, Davis, Short, & Webb, 2015; Vulkan et al., 2016). Notably, equity crowdfunding changes the dispersion of ownership from that of traditional venture investing: Equity crowdfunding shifts power to the entrepreneur by replacing a handful of larger outside investors with a multitude of many smaller ones. Both finance and management scholars have expressed concern about the trade-offs of ownership dispersion in a variety of organizational settings (e.g., access to a wider pool of financing but with greater agency costs). In our view, the dispersion of ownership via crowdfunding is likely to advance both theory and practice. Traditional agency theoretic approaches to corporate governance and incentive alignment may be challenged, with important implications for long-term versus short-term orientations, innovation behaviors, and capability development. Related management research explores how the role of structural embeddedness enables governance mechanisms to work to mitigate opportunism in complex network environments (Jones, Hesterly, & Borgatti, 1997). The role of structural embeddedness, however, is unclear in equity crowdfunding environments as the coalitions of funding agents are more ad hoc and temporary. Exploring alternative governance mechanisms in equity crowdfunding stands as an important avenue for future research.

A central component of crowdfunding is that it takes place in virtual settings. Prospective investors typically watch pitch videos and elect to invest over the Internet (Mollick, 2014). This raises important questions about the differences from and parallels to other classes of investors and how they evaluate opportunities. Many platforms explicitly utilize social proof or other signaling mechanisms to establish the legitimacy of new ventures. Since due

diligence can be difficult and limited in virtual environments, are there key differences in how these signals function, such as highly visible contagion effects or herding behaviors? From the perspective of the entrepreneur, what role do aspects of communication, such as video design and structure, play in shaping investor behavior? Theories of attentional control and communication will likely be useful lenses through which to view and understand such behaviors. In addition, new methodological approaches from fields such as information management (e.g., eye tracking software) may offer new windows into the underpinnings of visual dynamics and investment likelihood.

Furthermore, for equity crowdfunding investors, how do their motives affect the application of traditional theory related to portfolio diversification and risk management? To this point, crowdfunding tends to come from funders who are within the network of those seeking funding. As equity crowdfunding grows, how will the increase of investors with more traditional financial motives alter these early norms? Moreover, given the newness of crowdfunding, the decision practices among crowdfunders are still opaque and probably evolving as less sophisticated investors move up this new learning curve.

Accelerators. The rapid emergence of accelerators also presents new research opportunities. Accelerators are an organizational innovation with several attractive features (Hathaway, 2016; Shane, 2016). The study of accelerators would be aided by research conducted at various levels. For example, at a more macrolevel, accelerators seem to exert a powerful regional impact: "We see a shift in funding for startups in accelerator-treated regions: more deals, more dollars, and more local investment groups. This applies to startups that attend the accelerator and those that do not" (Hochberg & Fehder, 2015: 1202). Further unpacking the positive and negative regional and economic impact of accelerators could be very beneficial.

Study of accelerators as organizational forms is needed as well. The structures and approaches most effective in influencing start-up performance, as well as the distinct decision processes that go into optimizing cohort selection, are important future avenues of research. Moreover, what are the metrics for success? How do inter-start-up dynamics (within cohorts) and networks influence start-up development and success? In addition, acceleration is unlikely to benefit all entrepreneurs equally. Drawing on human capital theory (cf. Dimov & Shepherd, 2005), one can plausibly theorize that variations across entrepreneurs will influence the efficacy of the acceleration process. For example, do less experienced entrepreneurs reap more performance-related benefits from accelerators than their more experienced counterparts?

The acceleration process offers a unique window into the process of entrepreneurial learning. Entrepreneurial learning scholars have noted the need for a much better understanding of the dynamic relationships between mind, environment, and entrepreneurial action and have called for research to probe the interactions of different variables of cognitive interest across levels of analysis (Grégoire, Corbett, & McMullen, 2011). Yet collecting data for such research often proves difficult. Accelerators offer an accessible, natural lab setting to study entrepreneurial learning in action.

Directions for Future Research Across Forms of Financing

Venture financing methodology. The study of venture financing calls for a fundamental reevaluation of theory and research methodology. Venture investments do not often evenly

yield significant investment returns, and data suggest only a few investments generate the lion's share of industry returns (e.g., Cochrane, 2005). Understanding outcomes of entrepreneurial financing, then, revolves around identifying rare events and events that have extreme outcomes. The widespread use of correlational methods designed to estimate "average returns" can, however, yield misleading results since the average across a sample is influenced strongly by the extreme outliers. Whereas the tendency to remove extreme outliers in ordinary least squares models might aid researchers in finding significant results, much of the information of interest to equity financing researchers is represented by the outliers. Altering both our conceptual models and methodological tools to account for extreme outcomes is an important task for future research.

There have been multiple calls for scholars to consider the distributions of the events they are studying in order to better model and draw conclusions from their work (e.g., Sornett & Ouillon, 2012; Starbuck, 2016), considering, for example, power laws or Pareto distributions when examining rare and impactful events. With such an approach, scholars studying investment decisions, for example, may change how they view investment decision making—becoming less about the criteria that maximize returns of one opportunity and more about the evaluative strategies used to realize a small handful of wins in the midst of a larger, inevitable tranche of losses. The notion of affordable losses in effectuation theory (Dew, Read, Sarasvathy, & Wiltbank, 2009) might provide useful conceptual tools for understanding investor decision making. Adopting such an approach, versus a focus on isolated investment criteria, challenges and has the potential to advance theory that can improve empirical modeling and our understanding of VC investment activities.

Interconnectedness among funding sources. The growth of funding alternatives has several implications when considered alongside one another. Most studies consider funding sources in isolation, but as the landscape evolves, considering their interconnectedness becomes increasingly important. That is, shifting research designs and theoretical perspectives from an isolated focus on one mechanism to considering the interrelated or simultaneous presence of others stands as an important shift going forward. Entrepreneurs have historically had few options for high-growth funding, but their increasing choice set has important implications: The growing options may give more power to entrepreneurs to select, negotiate, and manage ongoing relations with investors. How the entrepreneur's expanding choice set challenges and changes traditional assumptions, dynamics, and theories remains an understudied area of inquiry.

Moreover, the emergence of different entrepreneurial funding sources presents situations in which participants can be friends and/or foes (cf. Hellmann & Thiele, 2015). We view it as an important undertaking to deconstruct further the nuances of these relational dynamics. For example, regarding complementary relations (e.g., friends), some extant work suggests that accelerator activity increases subsequent VC activity (Hochberg & Fehder, 2015), and higher volumes of reward-based crowdfunding investors or the investment of a reputable angel group can result in more favorable VC evaluations (Drover, Wood, & Zacharakis, in press). Such findings begin to shed light on the various ways in which funding sources may work together, potentially serving as new avenues of deal flow and certification for later-stage investors. Thus, research is needed to identify how earlier-stage funding mechanisms communicate positive signals to other prospective investors and stakeholders (Ahlers et al.,

2015). On the other hand, in regard to adversary relationships (or foes), other work suggests funding models may compete or challenge one another: “The data show that equity crowdfunding will likely pose great challenges to VC and business angel financiers in the near future” (Vulkan et al., 2016: 37). The evolving relations are complex and conditional, but the implications within and across funding models introduce an important frontier; central to understanding the modern funding landscape is its increasingly interconnected nature. Emerging research on co-opetition in strategy (e.g., Gnyawali & Park, 2011) may prove useful in understanding the evolving equity financing process.

Concluding Remarks

The capitalization of new ventures is one of the most foundational issues of entrepreneurship and the launch of high-growth ventures. In a manner consistent with the variability and importance of attaining funding, an impressive number of studies have explored a multiplicity of aspects and outcomes of venture financing over the past several decades. While much of this work has focused on VC, models from this work are increasingly spilling over into the study of angel investment, CVC, crowdfunding, and accelerators. Together, this body of work is growing, and we see many opportunities for future work. This review has brought attention to the largely undersocialized perspective represented across this literature, and we encourage researchers to identify and study important sources of heterogeneity among equity investors, as well as important interactions across these different categories of investors. Emerging forms of equity financing, such as crowdfunding and accelerators, underscore the criticality of broadening our scope of inquiry to understand the entrepreneurial financing landscape in its entirety—parceling out where established theory is still relevant and where new theory needs to be forged. Going forward, careful theorizing accompanied by well-designed empirical studies is essential to probing how changes in practice and our theoretical understanding of this domain can coevolve.

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